GEORGE ZHANG

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EDUCATION

University of Massachusetts - Amherst **September 2025 - May 2026**
Master's, Computer Science *GPA: 3.94*
Relevant Coursework: Reinforcement Learning, Artificial Intelligence, Information Retrieval, Distributed & Operating Systems, Systems for Data Science

University of Massachusetts - Amherst **February 2023 - May 2025**
Bachelor's, Computer Science *GPA: 3.94*
Relevant Coursework: Machine Learning, Algorithms for Data Science, Software Engineering

PROFESSIONAL EXPERIENCE

2Witech Solutions **September 2025 - December 2025**
Application Developer *Remote*
- Contributed to the development and extension of an API supporting real-time detection and analysis of PFAS contaminants in water
- Assisted in debugging, testing, and validating data pipelines for reliability

PROJECTS & OUTSIDE EXPERIENCE

Multi-Agent Poker AI Framework | Python, Reinforcement Learning **March 2026 - May 2026**
- Developed a modular AI agent framework for imperfect-information Texas Hold'em gameplay under uncertainty
- Implemented belief-based decision agents using Monte Carlo win-probability estimation, heuristic expected-value modeling, and opponent behavior tracking
- Built Information Set Monte Carlo Tree Search (ISMCTS) agents for adversarial search in partially observable environments
- Designed self-play training pipelines and heuristic weight optimization systems for iterative policy improvement
- Optimized action-selection under constrained betting and real-time decision-making environments

ScaleRL Reinforcement Learning Platform | Python, PyTorch, Gymnasium, Stable-Baselines3 **November 2025 - December 2025**
- Developed a reinforcement learning framework for optimizing production scheduling under non-stationary demand conditions
- Designed a custom Gymnasium environment modeling inventory freshness, expiration, and time-dependent customer demand
- Implemented and evaluated PPO, A2C, and DQN agents
- Built evaluation pipelines and benchmarking tools to compare RL agents against heuristic baseline strategies
- Generated learning curves and operational metrics to analyze policy efficiency, inventory waste, and lost sales
- *Link to project*

Robustness Analysis of Biomedical Retrieval Systems | Python, LLMs, Information Retrieval **October 2025 - December 2025**
- Conducted empirical research on robustness of biomedical retrieval systems against LLM-generated adversarial hard-negative samples across several biomedical datasets
- Developed a two-stage synthetic data generation pipeline using GPT-4o to generate query-targeted hard negatives for retrieval robustness evaluation
- Implemented hard-negative injection and benchmarking workflows to measure retrieval degradation under adversarial distractor pressure across varying injection levels
- Evaluated BM25, BGE, MedCPT, and Contriever retrieval models under controlled corpus perturbations using NDCG, MAP, Precision@K, Recall@K, and custom robustness-oriented metrics

SKILLS

Skills: Python, Java, C/C++, JavaScript, TypeScript, SQL, Pytorch, Reinforcement Learning, Machine Learning, Docker, REST APIs, MongoDB, Postgres, React, Node.js, Express.js, Git
Languages: Mandarin